Software Construction and Development – Lab Mid Exam

(Open Internet Lab)

Time: 80 minutes | **Format:** Single .py file , named `NAME_FATHERNAME.py` | **Submission:** GCR

Topic: Digital Detoxification

Description A system that helps people reduce unhealthy screen time and regain focus and balance in daily life. It tracks usage habits, sets limits, and encourages offline activities. The goal is mindful technology use, not complete disconnection.

Question Build a useful, working Python program related to **Digital Detoxification**. You decide the idea (e.g., screen-time tracker, detox challenge planner, focus session manager, habit streak tracker).

Your single `.py` file must show what we learned in this course:

1. Modular design: classes and functions with clear single responsibilities.
2. Use **regex** to validate or extract data (e.g., time formats, app names, durations, or log lines).
3. Use **XML** to export or import at least one report or data set (e.g., a weekly usage summary).
4. Input validation and proper error handling (e.g., bad input, missing or corrupt files).
5. Meaningful names, docstrings or comments, and no hard-coded magic values.
6. A `main()` entry point and a short header comment explaining your design decisions.
7. Make sure you understand what you write else it is useless.

Rules: Open internet is allowed, but the code must be your own. Submissions are checked for similarity using software. Submit on GCR before time ends.

Marking will be relative, so helping each other will make sure that you get worst marks ☺